

ADITYA HANSOTY

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EDUCATION

University of Michigan - Ann Arbor

Aug 2025 - May 2029

Bachelor of Science in Biomedical Engineering, Minor in Computer Science

- GPA: 3.73/4.0

WORK EXPERIENCE

Prosthetics Research Intern

Chicago, IL

Rehabilitation Robotics Lab, University of Illinois Chicago

May 2026 - Present

- Designed threaded encoder caps for a powered ankle-foot prosthesis in SolidWorks, iterating tolerances across 3D-printed and CNC-machined revisions for device integration.
- Operated a powered ankle-foot prosthesis via TwinCAT and led Vicon motion-capture setup and calibration across 10 human-subject trials; developed Python pipelines to synchronize EMG, ECG, and metabolic-cost data for a journal manuscript.
- Engineered and open-sourced a signal-agnostic AR feedback platform (Python, Unity, HoloLens 2) that pixel-streams LSL biosignals to a head-mounted display; sole developer.
- Conducted a validation study using OpenCV marker tracking and regression across 10 clips at three injected delay levels; found on-device AR latency statistically indistinguishable from a monitor baseline (0 ± 10 ms), then ran a 2-subject pilot evaluating gait symmetry response to real-time AR feedback under induced asymmetry; first-author paper submitted to ICRA 2027.

Cancer Biology Research Intern

Chicago, IL

University of Chicago Medicine

Jun 2025- Aug 2025

- Investigated beta-tubulin isotype knockouts in bone-metastatic breast cancer cell lines, quantifying effects on cell adhesion and cytoskeletal organization across three experimental conditions.
- Characterized protein expression and cellular morphology using mammalian cell culture, adhesion assays, Western blotting, and confocal microscopy.
- Synthesized experimental findings into a scientific poster selected for presentation at the University of Chicago Comprehensive Cancer Center Symposium, communicating results to 100+ researchers.

ACTIVITIES

Biomedical Engineering Society

Ann Arbor, MI

Professional Development Executive Board

Aug 2025 - Present

- Led 10+ technical workshops for 100+ members on SolidWorks, wet-lab techniques, and resume development, coordinating instructors, curriculum, and materials throughout the academic year.

PROJECTS

EMG-Controlled Prosthetic Gripper

May 2025 - Present

- Designed and 3D-printed a 2-finger gripper in SolidWorks, integrating an MG996R servo, MyoWare 2.0 EMG sensor, and Arduino Uno with external power regulation to eliminate servo stutter under load.
- Benchmarked two EMG classification pipelines across electrode-replacement sessions; achieved 99.4% accuracy across three forearm gestures using linear discriminant analysis (LDA).

SKILLS

Engineering: SolidWorks, 3D Printing, CNC Machining, Arduino, Simulink, TwinCAT, Vicon Motion Capture, EMG/ECG Signal Processing, LSL

Programming: Python, C++, MATLAB, R, Unity, OpenCV

Data & ML: NumPy, Pandas, SciPy, scikit-learn